

Magnetic Equivalent Circuit Modeling of Yokeless Axial Flux Permanent Magnet Machine With Segmented Armature

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This paper presents a detailed magnetic equivalent circuit (MEC) to improve the analysis accuracy and reduce computation time in modeling of axial flux permanent magnet machines. In the proposed MEC model, the magnetic saturation, armature reaction, leakage flux, and rotor rotation are considered to calculate the machine's static characteristics. The investigated machine structure and feature are first introduced. Then, the air-gap flux density distribution, back electromotive force waveform, and the average torque are calculated with the proposed method. Finally, the accuracy of the proposed method is verified by the 3-D finite-element method, and good agreements are achieved.

Index Terms—Axial flux permanent magnet (AFPM) machine, finite-element method (FEM), magnetic equivalent circuit (MEC), static characteristic.

I. INTRODUCTION

AXIAL FLUX permanent magnet (AFPM) machines with pancake shape, compact construction, high efficiency, and high power density have gained increasing interests of many researchers [1]. AFPM machines are particularly suitable for electrical vehicles, pumps, machine tools, renewable energy systems, and industrial equipment wherever extreme axial compactness and high torque density are significant.

Due to the construction and geometry, AFPM machines represent 3-D electromagnetic problems, which increased the difficulty of the design, analysis, and optimization process. However, the machine's static characteristics, such as air-gap flux density distribution, back electromotive force (EMF), and average torque, are critical in machine analysis and design. Different methods are commonly used for the calculation, including analytical methods, the finite-element method (FEM), and the magnetic equivalent circuit (MEC) method.

Analytical methods predict the magnetic fields in a machine by solving Maxwell equations under some specific assumptions. However, oversimplifications in models, such as ignoring the slotting effect, magnetic saturation of iron core, and magnetic flux leakage, decrease the accuracy of the analysis results [2]–[4]. The FEM is widely used due to the advantage of high accuracy, but it is time consuming and requires very large amounts of computer memory, and can hardly be used in machine initial design procedures with several parameters change in wide ranges [5], [6].

The MEC method is considered to be a good compromise between computation time and accuracy. It gives the possibility to consider the magnetic saturation, armature reaction, leakage flux, and the rotor movement [7]. In this paper, a relatively simple and accurate MEC is proposed for the AFPM machine shown in Fig. 1. The investigated machine is introduced first, and then an MEC model of the AFPM machine is proposed

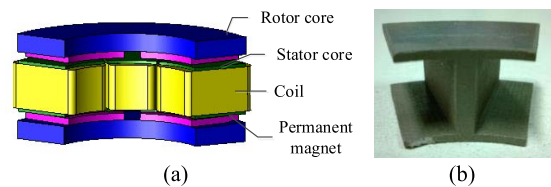


Fig. 1. (a) Construction of investigated machine. (b) SMC stator core.

to calculate the machine performance. All the results obtained by the method are validated by finite-element calculation, and good agreements are achieved.

II. ANALYSIS MODEL

Fig. 1(a) shows the construction of the test machine used in this paper. The AFPM machine consists of two rotors and one segmented stator. Soft magnetic composite (SMC) is used for the stator core, as shown in Fig. 1(b). Axially magnetized NdFeB permanent magnets are mounted on the surface of each rotor, which is composed of #45 steel. Three phase windings are concentrated on each stator segment to avoid overlapping. This machine topology benefits from many advantages, including short end windings with high winding fill factor, small stator iron loss in high-speed operation and convenient for manufacturing and assembly, reduced stator weight due to the absence of the stator yoke. Dimensions and specifications of the investigated machine are shown in Table I.

III. MEC

The MEC method has been improved steadily in recent years, and is very attractive for the design and optimization of different types of electrical machines, including induction machines [8], linear machines [9], AFPM machines [10], and so on. The MEC approach uses the reluctances, which are either time dependent or position dependent to model different machine parts to achieve high-accuracy calculation.

By assuming that the machine's static characteristic can be calculated at the average diameter, the AFPM machine is simplified to a simple 2-D model, as shown in Fig. 2. Only

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TABLE I
SPECIFICATIONS OF THE TEST MACHINE

Parameter	Symbol	Value
Rated Power (kW)	P	30
Rated voltage (V)	U	400
Stator outer diameter (mm)	D_o	174
Stator inner diameter (mm)	D_i	100
Air gap length (mm)	g	1.5
Stator axial height (mm)	h_s	34
Rotor axial height (mm)	h_r	15
Magnet height (mm)	h_m	4.5
Magnet pole arc ratio	-	0.8
Slot width (mm)	L_s	16
Shoe width (mm)	L_{sh}	7

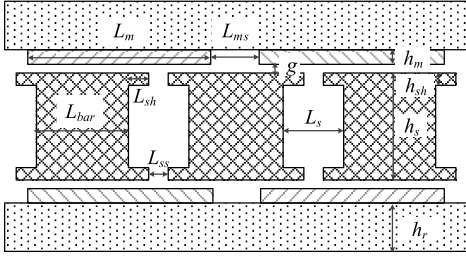


Fig. 2. Schematic view and symbols of the dimension of the machine.

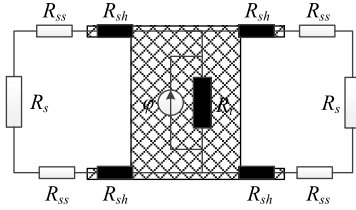


Fig. 3. MEC for a single stator module.

one of the three pole pairs is modeled and a periodic boundary condition is used because of the symmetry. The MEC model consists of four main parts: 1) the rotor core; 2) the magnet; 3) the air gap; and 4) the stator core.

Fig. 3 shows the MEC of a single stator module. Ferro-magnetic materials parts are modeled by nonlinear reluctance R_t and R_{sh} , which are painted in black. A flux source φ is introduced to consider the armature reaction. Leakage fluxes passing through slots are modeled by the reluctance R_s . Leakage fluxes between two adjacent shoes are considered by the reluctance R_{ss} . The values of these reluctances can be calculated by

$$R_t = \frac{h_s}{\mu_0 \mu_r(B) L_{bar} L_{act}} \quad (1)$$

$$R_{sh} = \frac{L_{sh}}{\mu_0 \mu_r(B) h_{sh} L_{act}} \quad (2)$$

$$R_s = \frac{h_s - 2h_{sh}}{\mu_0 L_s L_{act}} \quad (3)$$

$$R_{ss} = \frac{L_{ss}}{\mu_0 h_{sh} L_{act}} \quad (4)$$

where L_{act} represents the active length of the AFPM machine, which is the difference between the outer radius and inner

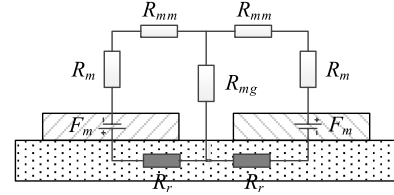


Fig. 4. MEC for permanent magnet and rotor core.

radius, and h_s , L_{bar} , L_{sh} , h_{sh} , L_s , and L_{ss} denote the stator axial height, the bar length, the shoe length, the shoe height, the slot width, and the distance between two adjacent shoes, respectively, as shown in Fig. 2. Relative permeability of the stator core $\mu_r(B)$ varies with the flux density, and it can be taken from the $B-H$ curve of the SMC material with an iteration process to consider the saturation.

The magnet is modeled by considering the air-gap and zigzag leakage fluxes [11], including the equivalent magnetomotive force source F_m , the magnet reluctance R_m , the magnet to magnet leakage flux reluctance R_{mm} , and the reluctance corresponding to magnet to air-gap leakage flux R_{mg} , as shown in Fig. 4. Similarly, the rotor core is modeled by nonlinear reluctance R_r to consider the magnetic saturation.

Above reluctances and magnetomotive force can be calculated as follows:

$$F_m = H_c h_m = \frac{B_r h_m}{\mu_0 \mu_r} \quad (5)$$

$$R_m = \frac{h_m}{\mu_0 \mu_r L_m L_{act}} \quad (6)$$

$$R_{mm} = \frac{\pi}{\mu_0 L_{act} \ln \left(1 + \frac{\pi g}{L_{ms}} \right)} \quad (7)$$

$$R_{mg} = \frac{h_m}{\mu_0 L_{ms} L_{act}} \quad (8)$$

$$R_r = \frac{L_m + L_{ms}}{\mu_0 \mu_r(B) h_r L_{act}} \quad (9)$$

where H_c , μ_r , and B_r are coercive force, relative permeability, and remanence of the permanent magnet, respectively. h_m and L_m represent permanent magnet height and length, and L_{ms} denote the distance between two adjacent magnets.

The air gap is divided into several independent elements according to magnets and the space between adjacent magnets, together with the slots and shoes. As shown in Fig. 5, when the rotor rotates, the values and numbers of air-gap reluctances and the connectivity of the reluctances change with regard to the position of the stator and magnets. The air-gap flux path is considered to be axially and perpendicular to the rotor with acceptable accuracy. The air-gap reluctance of each divided elements is given by

$$R_i = \frac{g}{\mu_0 A_i} \quad (10)$$

where A_i are air-gap areas that perpendicular to the air-gap flux path of every divided elements.

Five time steps are taken to obtain the machine performance in a quarter pole pitch, and then the antiperiodic and periodic conditions are used. Fig. 6 shows the MEC model of the first

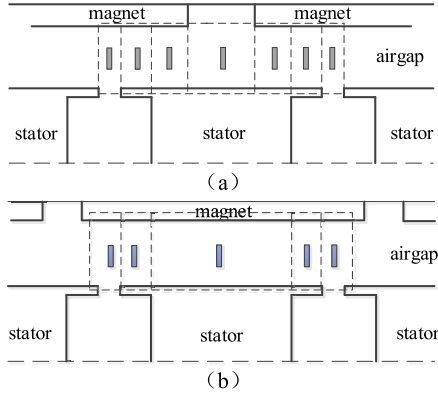


Fig. 5. (a) Magnet face to stator slot. (b) Magnet face to stator tooth.

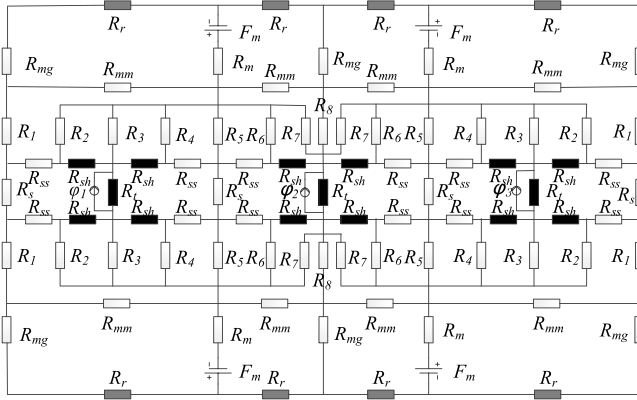


Fig. 6. Complete MEC for the investigated machine.

TABLE II
CALCULATED RELUCTANCES OF ALL PARTS

Symbol	Value (μH^{-1})	Symbol	Value (μH^{-1})	Symbol	Value (μH^{-1})
R_t	varied	R_{mg}	6.740	R_4	4.609
R_s	4.033	R_{mm}	133.9	R_5	16.13
R_{sh}	varied	R_1	2.247	R_6	4.609
R_{ss}	10.75	R_2	3.934	R_7	3.695
R_m	1.623	R_3	1.014	R_8	2.247
R_r	varied				

time step, as shown in Fig. 5(a) with different parts combined together. The node pairs on the left and the right lines of the MEC are set as equipotential nodes to consider the periodic boundary. The relative reluctances of Fig. 6 are calculated and listed in Table II.

IV. SOLVING METHOD

To solve the MEC model, a lumped-parameter electrical network approach that uses the electrical circuit analogy is put forward, where reluctances are modeled as resistances and the magnetomotive forces are treated as electric voltage. In this way, the magnetic circuit is transformed into a nonlinear electrical circuit that contains the iterative process to consider the magnetic saturation.

Based on above ideas, a transient simulation model is achieved by MATLAB/Simulink, and the magnetomotive force and flux of each element that is analogized by voltage and

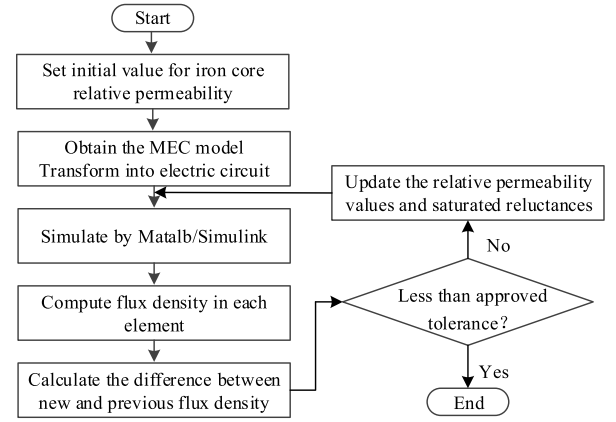


Fig. 7. Operation process for solving the MEC model.

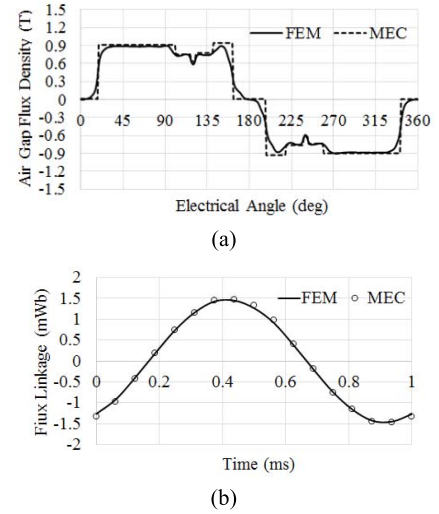


Fig. 8. (a) Air-gap flux density distributions. (b) Flux linkage distributions.

current, can be obtained in a simple and convenient way for further analysis. The operation process for solving the MEC model of the investigated machine is shown in Fig. 7.

V. MACHINE PERFORMANCE CALCULATION

In this section, machine characteristics are calculated by both the proposed MEC model and 3-D FEM to verify the accuracy of the proposed method.

A. Air-Gap Flux Density Distribution

Fig. 8 compares the calculated air-gap flux density and flux linkage waveforms obtained by the proposed method and 3-D FEM under the no-load condition. Good accordance is achieved for both waveforms.

B. Back EMF

The back EMF is one of the most significant characteristics for the machine design and analysis. It can be evaluated from the air-gap flux linkage distribution as

$$E_{ph} = -N_{ph} \frac{\Delta \Phi}{\Delta t} \quad (11)$$

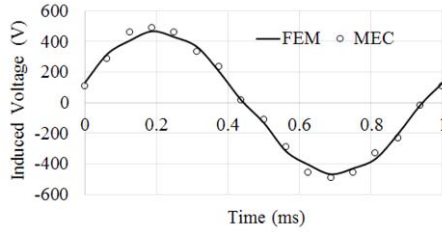


Fig. 9. No-load phase voltage waveforms obtained by the proposed method and 3-D FEM.

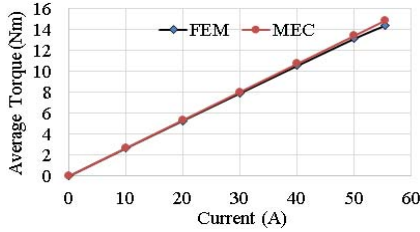


Fig. 10. Variations of average torque with stator current obtained by the proposed method and 3-D FEM.

where N_{ph} is the number of coil turns per phase and Φ is the air-gap flux linkage. The comparison between the back EMF waveform obtained by the proposed method and that of 3-D FEM is shown in Fig. 9. Compared with the 3-D FEM calculated results, the amplitude of the no-load voltage obtained by MEC model is $\sim 5\%$ higher, which is due to the assumption that the air-gap flux lines are always axially oriented.

C. Average Torque

Unlike cylindrical machines, the electromagnetic torque of AFPM machines is produced over a continuum of radii. By assuming that the torque can be calculated at the average diameter, the average electromagnetic torque can be simply calculated from

$$T = \frac{3}{\sqrt{2}} p N_{ph} k_w \Phi_f I \quad (12)$$

where p is the number of pole pairs, k_w is the winding factor, Φ_f is the magnetic flux excited by permanent magnet per pole, and I is the stator current. As shown in Fig. 10, the results of the proposed method are in good accordance with the simulation of 3-D FEM.

D. Computation Time Comparison

The 3-D FEM is the most accurate numerical method used for AFPM machines analysis, which considers the 3-D effects in the machine electromagnetic behavior. However, the preparation time needed to build up the 3-D simulation model and computation time to solve it is extremely large. The consumed time of the proposed MEC method is significantly reduced compared with the FEM, as shown in Table III. Compared with the 3-D FEM, the preparation time of the proposed method, which spends in making circuit models and writing a program to calculate the reluctances values of each step, is reduced by $\sim 50\%$, while the computation time spend in solving the analogy nonlinear electrical circuit in each step and

TABLE III
COMPUTATION TIME OF FEA AND PROPOSED MODEL

Model	Numbers of nodes	Preparation time	Computation time
FEM	8448	About 1 hour	About 30 min
MEC	50-52	About 30 min	About 1 min

processing the calculation data with postprocessing programs is as short as $\sim 1/30$.

VI. CONCLUSION

This paper has presented a relatively simple and accurate MEC model for yokeless and segmented armature AFPM machine to calculate the machine's static characteristic. The model is able to consider the magnetic saturation, armature reaction, leakage flux, and the relative motion. The simulation results of the proposed method that are in good accordance with the 3-D FEM confirm the effectiveness of the MEC method. Moreover, the computation time is reduced to a large extent compared with the 3-D FEM.

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